Jonathan Chen

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Education

Purdue University West Lafayette, IN

Bachelor of Science in Data Science & Artificial Intelligence *Expected Graduation: May 2029*

- **Honors:** Lilly Scholar (Full Tuition Merit Scholarship), Coca-Cola Scholar Finalist, National Merit Scholar
- **Activities:** Designer @ Hack the Future, Lilly Purdue Innovation Institute, The Data Mine, Cornerstone Christian Fellowship, Toastmasters
- **Relevant Coursework:** Data Engineering, Multivariate Calculus, Discrete Mathematics, Python Programming, Object-Oriented Programming, Plane Analytic Geometry, Statistics

Experience

Walmart Global Tech Bentonville, AR

Technical Program Management Intern *January 2026 – April 2026*

- Developed a full-stack tracking dashboard (React, TypeScript, Python) that centralized project intake and dependency tracking across Walmart's global portfolio, integrating Jira and Confluence REST APIs
- Evaluated internal TPM tooling including Jira/Confluence MCP workflows, and AI agents for development (CodePuppy/Wibey)

Institute for Systems Biology (ISB), Baliga Lab Seattle, WA

Computational Biology Research Intern *June 2025 – August 2025*

- Developed R pipelines to analyze high-throughput phenotypic microarray and drug-response data; identified key metabolic vulnerabilities in 20 CRISPRi strains of *Mycobacterium tuberculosis*, informing drug regimen development
- Presented research findings to institute; assisted K-12 STEM curriculum development with the Systems Education Experiences team

City of Bentonville & Industry Partners Bentonville, AR

Software Engineer Intern *September 2025 – April 2026*

- Contracted by the City of Bentonville to engineer predictive traffic machine learning models (Python, Scikit-learn, Pandas, NumPy) using months of municipal time-series sensor data, reducing peak delays by 50%
- Architected an enterprise RFID-based tracking system to enhance product visibility across Walmart stores, presented to Walmart and Zebra Technologies executives

Sun Yat-sen University, Li-Meng Feng Lab Guangzhou, China

Wet Lab Research Assistant *May 2025 – June 2025*

- Generated primary data and performed RT-qPCR, Western Blot, DNA/RNA extraction, gel electrophoresis, and X-ray film imaging to support cancer research team

Etsy: LotusandBambooCrafts 2023 – Present

Small Business Co-Founder

- Manage personalized oriental-themed gift store; Etsy Star Seller status; generated $9,000+ profit from 600+ sales

Projects

Radically | *KickStarter Launch*

- Contracted to design and develop a novel Chinese learning web app (iF Design Award, TEDx), implementing 10+ interactive learning modules with centralized state management and persistent progress tracking (React, TypeScript, Zustand, Framer Motion, Supabase)
- Developed reusable, responsive UI components from Figma/Paper Design prototypes, integrating animated interactions and iterating on learning workflows through user testing

GuitarCade | *HogHacks 1st Place*

- Developed an acoustic-fingerprinting algorithm for a gamified guitar-learning web app, calibrating to each instrument's unique spectral profile to trigger game actions from real-time audio
- Used FFT/chroma analysis and cosine similarity with Python, NumPy, and PyAudio; integrated a React/Three.js interface through WebSockets and the Web Audio API

Technical Skills

Languages: Python, R, SQL, Java, JavaScript, TypeScript, HTML/CSS

Frameworks/Libraries: React, Next.js, Scikit-learn, OpenCV, NumPy, Pandas, tidyverse, ggplot2, Three.js

Tools: Git/GitHub, WebSockets, Supabase, Agile (Jira, Confluence), MCP Workflows, Power BI

Product/Design: Figma, Paper.design, UI/UX Wireframing, Adobe Creative Cloud, DaVinci Resolve